

The wrong co-transmitter: purinergic rather than peptidergic dominance as a mechanism for hard flaccid syndrome

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Seventh in a series. Companion papers: *The sacral splanchnic bypass*; *Initiation without maintenance*; *The intraganglionic loop*; *A generator that outlives its cause*; *Autonomy without a signal*; *An amplifier without a signal*.

Abstract

Background. Sympathetic nerves do not release noradrenaline alone. If the contractile signal in hard flaccid syndrome (HFS) has shifted away from noradrenaline toward a co-transmitter, the resulting contraction would be incompletely reversible by α -adrenoceptor blockade — which is what patients and clinicians report.

The conjecture, and why it names the wrong molecule. The conjecture as proposed identifies neuropeptide Y (NPY) as the co-transmitter responsible. The penile-specific evidence does not support this. In penile small arteries NPY acts as a *dual modulator*, potentiating noradrenaline-evoked contraction through Y1 and Y2 receptors while producing nitric-oxide-independent relaxation through an atypical endothelial receptor. Burnstock's canonical account is that NPY, when released, acts largely as a postjunctional neuromodulator potentiating the responses to noradrenaline and ATP, and as a prejunctional modulator of their release. **A transmitter whose contractile action is potentiation of noradrenaline cannot survive α -blockade, because α -blockade removes what it potentiates.**

The reformulation. ATP is the genuine independent sympathetic co-transmitter. It is co-stored with noradrenaline, acts at postjunctional P2X1 receptors to produce depolarisation, calcium influx, calcium sensitisation and contraction, and is therefore intrinsically resistant to α -blockade. P2X1 receptors are expressed on corpus cavernosum smooth muscle and mediate part of the nerve-evoked contraction. We restate the conjecture as *purinergic* rather than *peptidergic* dominance.

A second correction, and a way out. The conjecture's frequency argument — that sustained high-frequency firing favours co-transmitter release — holds for NPY but is reversed for ATP: short low-frequency bursts favour ATP while prolonged stimulation favours noradrenaline. The release-pattern route to purinergic dominance is therefore blocked. We propose a receptor-level route instead, for which there is a documented human precedent in genitourinary smooth muscle: the purinergic component of human detrusor contraction rises from roughly 3–5% to around 40% in pathological states and increases with age.

An overlooked mechanism. α -adrenoceptor blockade markedly enhances the release of both noradrenaline and NPY through loss of prejunctional $\alpha 2$ autoinhibition. Blocking α -receptors therefore removes one transmitter while increasing the release of the others — a mechanism for incomplete or paradoxical response that requires no chronic remodelling at all.

Conclusion. The conjecture's logic is sound and its molecule is wrong. Reformulated around ATP and receptor-level change, it becomes testable in an existing animal model, though the human pharmacological tools do not exist.

Keywords: hard flaccid syndrome; sympathetic co-transmission; ATP; P2X1 receptor; neuropeptide Y; corpus cavernosum; phentolamine; prejunctional $\alpha 2$ autoreceptor.

Reader advisory

This is hypothesis generation, not a report of new data. No component of the mechanism proposed here has been demonstrated in a person with hard flaccid syndrome. No purinergic receptor antagonist is licensed for human use for any indication discussed here, and none should be sought. Nothing in this paper should influence clinical management.

1. Introduction

Three papers in this series place the maintaining lesion in the effector tissue rather than in the nerve [3, 4], on the grounds that sympathetic blockade is reported to fail [E] [1, 2]. This paper examines a fourth route to the same conclusion, and one that does not require the effector to have changed at all: the possibility that the *transmitter* has changed.

Sympathetic postganglionic nerves release at least three substances. The concept of co-transmission was proposed by Burnstock in 1976 and is now well established, with a chemical coding for sympathetic nerves of noradrenaline, adenosine 5'-triphosphate (ATP) and neuropeptide Y (NPY) [A] [VERIFY] [10, 8]. If the balance among these has shifted in HFS, then α -adrenoceptor blockade — which targets only one of the three — would be expected to produce partial and variable relief. That is what is reported [E] [1], and it is the observation the conjecture was constructed to explain.

The conjecture is well-motivated. It is also, we will argue, built around the wrong molecule and the wrong mechanism of change — and it can be repaired on both counts, with the repaired version resting on better evidence than the original.

2. Methods

This is a hypothesis and theory article. No new human or animal data were generated; no patients were involved; ethical approval was not required.

Literature was identified by targeted searching of MEDLINE/PubMed and the open web between May and August 2026, using combinations of the terms *sympathetic cotransmission*, *neuropeptide Y penile artery*, *P2X1 corpus cavernosum*, *purinergic cotransmission ATP noradrenaline*, *frequency dependent transmitter release*, and *hard flaccid syndrome*, with backward citation chasing. The search was purposive and non-systematic. We searched specifically for the penile-tissue pharmacology of NPY, because the conjecture makes a claim about it, and Section 4 reports what that returned.

Claims are graded as in Table 1. Claims not verified against a retrievable source are marked [VERIFY]; quantities that do not exist in the literature are marked [DATA NEEDED].

Table 1. Evidence grading scheme. The grade refers to the strength of evidence for the *stated proposition*, not to its relevance to hard flaccid syndrome. No proposition here carries grade A evidence *in HFS*, because none exists.

Grade	Definition
[A]	Replicated human evidence, or a physiological principle established across independent laboratories and preparations.
[B]	Single human study, human tissue study, or human data with substantial methodological limitation.
[C]	Experimental animal or in vitro evidence, with species and preparation stated.
[D]	Mechanistic inference: deduced from A–C evidence but not itself directly observed.
[E]	Unverified anecdotal or community-sourced report; not peer reviewed, denominators unknown.

3. Background

3.1 Three transmitters, three roles

The roles are not symmetrical, and the asymmetry is the whole argument of this paper.

Noradrenaline acts at α 1-adrenoceptors, a metabotropic pathway producing inositol trisphosphate generation, calcium release and a slow contraction [A] [VERIFY] [10, 17].

ATP is co-stored with noradrenaline in sympathetic vesicles and, when co-released, acts at postjunctional P2X1 receptors to elicit depolarisation, calcium influx, calcium sensitisation and contraction [A] [VERIFY] [9, 8]. Because P2X1 is an ionotropic receptor with no adrenergic component, the resulting contraction is intrinsically unaffected by α -blockade [D].

NPY is stored in large granular vesicles and, when released, *acts largely as a postjunctional neuromodulator, potentiating the responses to both noradrenaline and ATP*, as well as acting as a prejunctional modulator of their release [A] [VERIFY] [8]. Its Y1 receptor is the predominant vascular subtype and mediates vasoconstrictor actions of the peptide in some beds [C] [VERIFY] [7].

3.2 The penile-specific evidence on NPY

This is where the conjecture runs into difficulty, and the relevant work is not obscure.

In penile small arteries, the sympathetic co-transmitter NPY *modulates* noradrenergic vasoconstriction — *both enhancing and depressing* noradrenaline contractions, through Y1 and Y2 postsynaptic receptors and through a nitric-oxide-independent atypical endothelial receptor respectively [C] [5]. The underlying study, in horse penile small arteries, reported a dual modulatory action: postjunctional potentiation of noradrenaline-induced contractions involving both Y1 and Y2 receptors, and relaxant effects mediated by a receptor with an atypical pharmacological profile probably located at the endothelium and unmasked in the presence of selective antagonists of the contractile receptors [C] [6].

Two features of that description matter. The contractile action is **potentiation of noradrenaline**, not independent contraction. And the net effect of NPY in this bed is **bidirectional**, with a relaxant arm that appears when the contractile receptors are blocked.

3.3 The penile-specific evidence on ATP

Corpus cavernosum smooth muscle cells of several species, including humans, express purinergic P2X receptors [C] [VERIFY] [13]. In mouse corpus cavernosum smooth muscle, isolated myocytes expressed P2X1 receptors and ATP evoked large inward currents (mean approximately 900 pA at -60 mV); α,β -methylene ATP produced a similar current; and P2X1-mediated contractions were recorded in whole tissue in response to electrical field stimulation [C] [13]. The authors state the obvious limitation themselves: the work used animal rather than human tissue.

The nearest human evidence is from an adjacent genital smooth muscle. In human vas deferens, electrical field stimulation-induced contractions were reduced to 40% of the initial contraction in the presence of a non-selective purinergic antagonist, indicating a functionally significant purinergic component acting most probably through P2X1 [B] [VERIFY] [14]. ATP has been shown to be a co-transmitter with noradrenaline in sympathetic nerves supplying the human vas deferens [B] [VERIFY] [8].

3.4 Frequency coding

The conjecture's dynamic argument rests on this literature, so the details matter.

For NPY, the conjecture is correct. In the blood-perfused pig spleen *in vivo*, splenic nerve stimulation with intermittent bursts at high frequency (20 Hz) caused a several-fold larger release of NPY-like immunoreactivity relative to noradrenaline than continuous stimulation at low frequency (2 Hz) delivering the same total number of impulses [C] [VERIFY] [11].

For ATP, it is reversed. The relative contribution of each transmitter depends on stimulation parameters: short bursts (1 s) at low frequency (2–5 Hz) favour ATP release, whereas longer periods of nerve stimulation (30 s or more) favour noradrenaline release [A] [VERIFY] [10].

4. Two corrections

4.1 Correction 1: NPY cannot do the job the conjecture assigns it

The conjecture states that under prolonged high-frequency firing the contractile signal shifts toward NPY, and that “constriction of this kind is characteristically resistant to α -adrenoceptor blockade” [1].

The second clause does not follow from the penile evidence. If NPY's postjunctional contractile action in this tissue is *potentiation of noradrenaline-evoked contraction* [6, 5, 8], then removing noradrenaline from the junction removes the substrate that NPY potentiates. α -blockade should abolish the NPY-potentiated component along with the noradrenergic one. Far from explaining α -blockade resistance, NPY dominance predicts α -blockade *sensitivity* [D].

The bidirectionality compounds the problem. NPY in penile small arteries also produces relaxation through an atypical endothelial receptor, an effect unmasked when the contractile receptors are blocked [6]. A shift toward NPY dominance could therefore reduce tone rather than raise it, depending on receptor complement. **The conjecture's prediction is not merely unsupported; its sign is uncertain.**

There is a form in which NPY could resist α -blockade — direct Y1-mediated vasoconstriction, which is documented in some vascular beds [7]. But the penile-specific characterisation is of modulation, not direct constriction, and it is the penile characterisation that is relevant here.

4.2 Correction 2: the frequency argument cannot deliver purinergic dominance

Suppose we substitute ATP for NPY, which Section 5 argues we should. The conjecture's dynamic mechanism then fails for the opposite reason. Prolonged sympathetic firing — the condition the conjecture posits — favours *noradrenaline*, not ATP [10]. The firing pattern that would favour purinergic transmission is short, low-frequency bursts, which is not what a tonically overactive sympathetic outflow provides.

So the conjecture faces a bind. **The co-transmitter whose release pattern fits its premise (NPY) is a modulator that α -blockade defeats; the co-transmitter that would genuinely resist α -blockade (ATP) is favoured by the opposite firing pattern.** Neither branch works as stated.

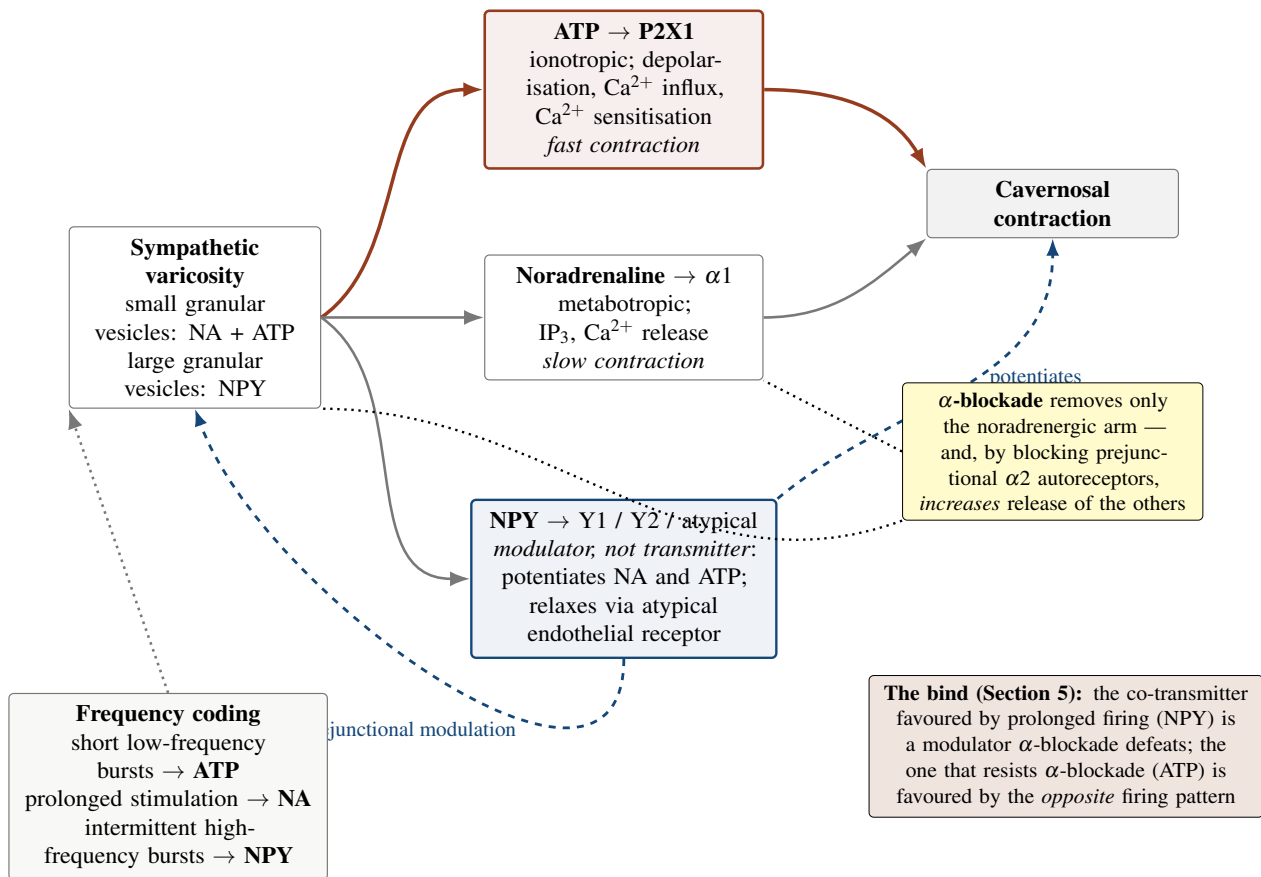


Figure 1. The sympathetic neuroeffector junction, the three transmitters, and why the conjecture’s logic does not close. ATP acting at ionotropic P2X1 receptors is the only one of the three whose contractile action is independent of adrenoceptors and therefore intrinsically resistant to α -blockade. NPY is a modulator: it potentiates the actions of noradrenaline and ATP and modulates their release, and in penile small arteries it also relaxes through an atypical endothelial receptor. The frequency-coding panel shows why a release-pattern account cannot deliver purinergic dominance under tonic sympathetic overactivity. Note also the prejunctional consequence of α -blockade, developed in Section 5.3. Diagram is schematic; the penile NPY pharmacology is from horse small arteries [6] and the P2X1 data from mouse cavernosal tissue [13].

5. The reformulation

5.1 Substitute the molecule

ATP is the sympathetic co-transmitter that can do what the conjecture requires. It is co-stored and co-released with noradrenaline; it acts through an ionotropic receptor with no adrenergic component; and the resulting contraction involves depolarisation, calcium influx and calcium sensitisation [9, 8]. P2X1 receptors are present on cavernosal smooth muscle and contribute to nerve-evoked contraction [13]. A shift toward purinergic dominance would produce exactly the phenotype the conjecture describes: a nerve-evoked contraction that α -blockade only partially reverses [D].

5.2 Substitute the mechanism of change

If the release-pattern route is blocked (Section 4), the shift must occur at the receptor. There is a documented human precedent for precisely this, in genitourinary smooth muscle, and it is the strongest single piece of support available for any version of this conjecture.

Upregulation of purinergic responses in human detrusor — from roughly 3–5% of the contractile response to around 40% — is well recognised in pathological conditions such as interstitial cystitis and overactive bladder, and the purinergic component also increases with ageing [B] [VERIFY] [13]. This is a nearly ten-fold change in the transmitter balance of a human visceral smooth muscle, occurring in disease, in an organ innervated by the same pelvic autonomic outflow that supplies the penis.

The therapeutic corollary in that field is instructive: $\beta 3$ -adrenergic agonists are relatively ineffective at relaxing cholinergic detrusor contractions but much more effective at suppressing the purinergic component [13]. A pathological shift toward purinergic transmission changes which drugs work — which is the clinical signature the conjecture is trying to explain.

5.3 An overlooked mechanism that needs no remodelling at all

There is a further possibility that the conjecture does not consider and that requires no chronic change of any kind.

In the pig spleen preparation, α -adrenoceptor blockade by phentolamine *markedly enhanced both noradrenaline and NPY release*, especially at low stimulation frequency, indicating prejunctional adrenergic inhibition of release; and after adding propranolol, a large remaining perfusion-pressure response to nerve stimulation was unmasked [C] [VERIFY] [11].

The implication is direct [D]. Non-selective α -blockade removes the noradrenergic contractile arm while simultaneously *increasing* the release of ATP and NPY, because prejunctional α_2 autoreceptors normally restrain the release of all three. A drug that subtracts one transmitter and adds to the others may produce little net change in tone — and might in principle worsen it. **This offers an explanation for incomplete or paradoxical phentolamine response that requires no co-transmitter dominance, no chronic remodelling and no disease-specific change at all**, and it applies equally to healthy tissue.

We regard this as the most useful thing in this paper, and note that it is a competitor to the conjecture rather than a support for it: if prejunctional disinhibition explains the phentolamine observation, the observation stops being evidence for a shifted transmitter balance.

6. The hypothesis

Formal statement, reformulated

H7 (purinergic dominance). In hard flaccid syndrome the balance of sympathetic co-transmission at the cavernosal neuroeffector junction has shifted toward ATP acting at postjunctional P2X1 receptors, principally through receptor-level upregulation rather than through altered release patterns. The resulting contraction is only partially reversible by α -adrenoceptor blockade.

H7-receptor (the defensible form). P2X1 receptor density or function on cavernosal smooth muscle is increased. Prediction: the purinergic fraction of nerve-evoked contraction is elevated; the fraction rises with disease duration; α -blocker responsiveness falls correspondingly.

H7-release (the original form, which we do not recommend). Altered firing patterns shift transmitter release toward co-transmitters. Blocked by the frequency evidence in Section 3.4.

H7-NPY (the form as originally proposed). NPY dominance produces α -blockade-resistant contraction. Contradicted by the penile pharmacology in Section 3.2.

H7-null. Co-transmitter balance in HFS is normal. Incomplete α -blocker response reflects prejunctional disinhibition (Section 5.3), inadequate dosing, or a mechanism elsewhere.

Auxiliary assumptions for the defensible form:

- **A1 (human relevance).** P2X1-mediated contraction is functionally significant in *human* cavernosal tissue. Demonstrated in mouse [13]; asserted for humans on the basis of receptor expression; the functional fraction in human corpus cavernosum has not been quantified [DATA NEEDED].
- **A2 (upregulation).** Denervation, injury or chronic sympathetic overactivity upregulates the cavernosal purinergic component. Documented for human detrusor in other pathologies [13]; never examined in cavernosal tissue [DATA NEEDED].
- **A3 (magnitude).** The shift is large enough to produce a symptomatic elevation of resting tone rather than merely altering the response to nerve stimulation. Untested, and note that the purinergic component is characteristically *phasic* — a fast transient — which is not obviously suited to maintaining a tonic contraction [D]. This is a substantial problem and we return to it.

7. Predictions and tests

7.1 The test that could be done now

Prediction P1. In cavernosal tissue from animals subjected to cavernous nerve injury or chronic sympathetic overactivity, the fraction of electrical field stimulation-evoked contraction abolished by a purinergic antagonist is greater than in sham controls.

This is a conventional organ-bath experiment in an established model using established pharmacology, of the kind already performed in mouse cavernosal tissue [13]. It requires no new technology and no human participants. **It is the only proposition in this paper that is currently testable, and nothing else here is worth doing until it has been.** If injured cavernosal tissue shows no increase in purinergic fraction, H7 fails at its first step.

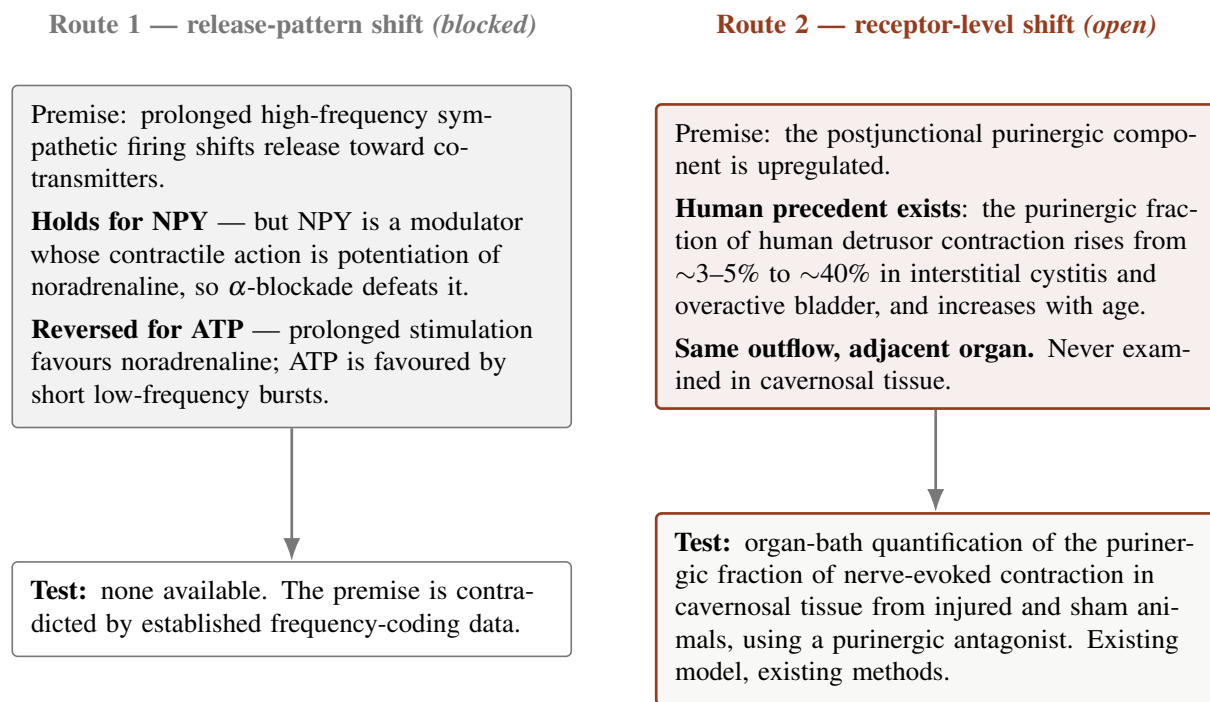


Figure 2. Two routes to co-transmitter dominance, and why we recommend the second. The release-pattern route is the one the conjecture proposes and is blocked by the frequency-coding evidence: the transmitter it favours is the wrong one and the transmitter that would work is disfavoured by the posited firing pattern. The receptor-level route is not blocked, has a documented human precedent in an adjacent organ supplied by the same pelvic autonomic outflow, and generates a test that could be performed in an existing animal model without new technology.

7.2 The free test, shared with two other conjectures

Prediction P2. α -blocker responsiveness declines with duration of illness.

The conjecture identifies this as its distinctive checkable prediction [1], and it is checkable retrospectively from existing records. We note honestly that it is not distinctive: calcium sensitization [3] and fibrotic remodelling [1] predict the same gradient, and a positive result would support a class of hypotheses rather than this one. It remains worth doing because nobody has done it and because the whole effector-autonomy family rests on a staging assumption that has never been examined.

7.3 The discriminating test, which cannot be performed in humans

Prediction P3. A P2X1 antagonist reduces resting cavernosal tone, and does so in patients in whom α -blockade has failed.

No purinergic receptor antagonist is licensed for human use for this or any comparable indication, and the research tools (suramin, PPADS, NF449 and related compounds) are non-selective, poorly characterised in humans, or both [VERIFY: WE DID NOT SYSTEMATICALLY REVIEW THE CURRENT CLINICAL DEVELOPMENT PIPELINE FOR P2X1-SELECTIVE ANTAGONISTS AND READERS SHOULD CHECK BEFORE CONCLUDING THAT NONE EXISTS]. The pharmacological discrimination the hypothesis needs is therefore unavailable, and we would not propose a human study.

7.4 Testing the alternative explanation

Prediction P4. If prejunctional disinhibition (Section 5.3) explains incomplete phentolamine response, then a selective α_1 antagonist — which does not block prejunctional α_2 autoreceptors and therefore does not disinhibit co-transmitter release — should outperform non-selective α -blockade at equivalent postjunctional blockade.

This is a clean and clinically feasible comparison, uses licensed drugs, and would separate a transmitter-balance explanation from a much more mundane one. It requires the objective tone endpoint that does not yet exist [2, 3] [DATA NEEDED]. We regard it as more informative than anything else proposed in this paper for the human setting.

7.5 The problem of phasic versus tonic

We flag one difficulty that applies to every version of H7 and that the conjecture does not address. The purinergic component of sympathetic co-transmission is characteristically the *fast, phasic* arm — ATP produces excitatory junction

potentials and a rapid transient contraction, while noradrenaline produces the slow, sustained component [10, 9]. Hard flaccid syndrome is a disorder of *sustained* tone. A shift toward the phasic transmitter is not an obvious route to a tonic contraction [D].

Two possible resolutions: P2X1 activation produces calcium sensitisation as well as calcium influx [9], which could outlast the transient; and receptor upregulation might alter desensitisation kinetics. Both are speculative, neither is demonstrated, and we regard this as the sharpest internal objection to the reformulated hypothesis.

Table 2. Predictions of H7 compared with the principal alternatives.

Observation	H7 (purinergic dominance)	Calcium sensitization	Prejunctional disinhibition	Discriminating value
Increased purinergic fraction of nerve-evoked contraction after injury	Predicted	Not predicted	Not predicted	High
α 1-selective blockade outperforms non-selective α -blockade	Not predicted	Not predicted	Predicted	High
Incomplete response to phentolamine	Predicted	Predicted	Predicted	None
α -blocker response declines with duration	Predicted	Predicted	Not predicted	Low
P2X1 antagonist reduces resting tone	Predicted	Not predicted	Not predicted	Decisive but unavailable
Tone is sustained rather than phasic	Not predicted	Predicted	Not applicable	Moderate, against H7
Y1 antagonist adds benefit to α -blockade	Predicted by the original NPY form only	Not predicted	Not predicted	Would discriminate the abandoned form

8. Alternative explanations

1. **Prejunctional disinhibition (Section 5.3).** The simplest explanation of incomplete non-selective α -blockade, requiring no disease-specific change. We regard it as more likely than H7 on current evidence.
2. **Inadequate dosing or delivery.** The phentolamine observation motivating the conjecture is unverified, undenominated and derived from uncontrolled use [E] [1]. Incomplete response to a drug at unknown dose is weak evidence for anything.
3. **Calcium sensitization.** RhoA/Rho-kinase upregulation would produce incomplete α -blocker response by a different route [3], and — unlike H7 — produces a sustained rather than phasic contraction, which fits the clinical picture better.
4. **Denervation supersensitivity.** Nonspecific supersensitivity would also produce partial and variable α -blocker response [4], and is distinguished from H7 by supersensitivity to unrelated agonists rather than by transmitter fraction.
5. **The premise may be false.** As in every paper in this series: cavernosal tone has never been objectively shown to be elevated in HFS, and the treatment-response reports that motivate the whole effector-autonomy family are grade E.

9. Implications

No clinical implications are drawn. Three research implications follow.

First, do the organ-bath experiment. Whether cavernosal purinergic transmission increases after injury is answerable now, in an existing model, with existing pharmacology. It is the only decisive step available.

Second, compare selective and non-selective α -blockade. If the phentolamine observation is an artefact of prejunctional disinhibition rather than evidence of transmitter shift, that is worth knowing before any further theory is built on it. The comparison uses licensed drugs.

Third, drop the NPY formulation. It is contradicted by the penile-specific pharmacology, and continuing to cite NPY as an α -blockade-resistant contractile transmitter in this tissue would propagate an error.

10. Limitations

1. **No evidence in HFS.** Co-transmitter balance has never been measured in a patient with this condition.
2. **The conjecture as originally stated is contradicted,** not merely unsupported, by the penile NPY pharmacology.
3. **The reformulation rests on mouse tissue.** The functional purinergic fraction in human corpus cavernosum has not been quantified.
4. **The phasic-tonic mismatch is unresolved** and is the sharpest internal objection to the reformulated hypothesis.

5. **The human precedent is analogical.** Detrusor purinergic upregulation is a striking parallel but a different organ, different innervation type (parasympathetic cholinergic co-transmission), and different pathologies.
6. **The discriminating pharmacology does not exist in humans.**
7. **The most useful mechanism identified here is a competitor.** Prejunctional disinhibition explains the motivating observation without the hypothesis.
8. **Several load-bearing citations are flagged unverified,** including the detrusor upregulation figures and the frequency-coding statements, which were traced through reviews and topic summaries rather than primary sources.
9. **Non-systematic search.** We did not systematically review the P2X1 antagonist development pipeline and our statement that no human tool exists should be checked.
10. **Confirmation risk.** This paper was written to develop a specific conjecture and ends by recommending that its central claim be abandoned and replaced. That is an unusual outcome and readers should satisfy themselves that the penile NPY pharmacology says what we report it saying before accepting it.

11. Refutation criteria

- **Substantially refuted.** Organ-bath studies show no increase in the purinergic fraction of nerve-evoked cavernosal contraction after nerve injury or chronic sympathetic stimulation relative to sham controls.
- **Substantially refuted.** α 1-selective blockade normalises resting tone in patients in whom non-selective α -blockade failed, indicating prejunctional disinhibition rather than transmitter shift.
- **Weakened.** α -blocker response shows no relationship to symptom duration.
- **Weakened further.** Human corpus cavernosum is shown to have a negligible functional purinergic component under any condition.
- **Already contradicted (NPY form).** The penile pharmacology of NPY is modulatory and bidirectional, not independently contractile.
- **Rendered moot.** Direct measurement shows resting cavernosal tone is not objectively elevated in symptomatic patients.

12. Conclusion

The reasoning behind this conjecture is better than its content. Sympathetic nerves do release more than noradrenaline; a shift in the balance would produce incomplete α -blockade; and that is a genuinely good explanation for a genuinely puzzling clinical report. But the molecule named cannot do the job. In penile small arteries neuropeptide Y is a dual modulator — potentiating noradrenaline through Y1 and Y2 receptors and relaxing through an atypical endothelial receptor — and Burnstock's canonical account of sympathetic co-transmission describes it as a postjunctional neuromodulator rather than an independent transmitter. A substance whose contractile action is the potentiation of noradrenaline cannot survive the removal of noradrenaline.

ATP can. It is co-released, acts at an ionotropic receptor with no adrenergic component, produces depolarisation and calcium sensitisation, and is expressed and functional on cavernosal smooth muscle. Reformulated around ATP, the conjecture becomes coherent — but only if the shift occurs at the receptor rather than in the release pattern, because prolonged sympathetic firing favours noradrenaline while ATP is favoured by short low-frequency bursts. That receptor-level reformulation has a real human precedent: the purinergic fraction of detrusor contraction rises roughly ten-fold in bladder disease and with age, in an organ supplied by the same pelvic outflow.

Two things then remain. One is a problem: the purinergic arm is the fast, phasic component of sympathetic transmission, and this is a disorder of sustained tone. The other is an alternative that may make the whole hypothesis unnecessary. α -adrenoceptor blockade markedly increases the release of the co-transmitters by removing prejunctional autoinhibition — so a non-selective α -blocker subtracts one transmitter and adds to the others, in healthy tissue as much as in diseased. If that is what the phentolamine reports reflect, then the observation this conjecture exists to explain is not evidence of anything having changed. Comparing a selective α 1 antagonist against a non-selective one would tell us, using drugs that already exist, whether there is anything here to explain at all.

Declarations

Authorship and the use of artificial intelligence. The second listed author is a large language model. It is listed at the corresponding author's request and in the interest of transparency, but this contravenes the recommendations of the International Committee

of Medical Journal Editors, which hold that artificial intelligence tools cannot meet authorship criteria because they cannot take responsibility for the accuracy and integrity of the work. **Before submission to any peer-reviewed venue this attribution should be converted into an AI-use disclosure, with the human author accepting sole responsibility for the content.** Because this paper argues that the conjecture's central claim is mistaken, the penile neuropeptide Y pharmacology on which that argument rests should be read in full and independently confirmed before submission.

Ethics approval. Not applicable. No human participants, animals or identifiable data were involved.

Funding. None.

Competing interests. No competing financial interests. The first author maintains a public website on hard flaccid syndrome and authored the source article [1]; this is a non-financial interest in the hypothesis being taken seriously, and readers should weigh the argument accordingly.

Data availability. No datasets were generated or analysed.

Clinical disclaimer. This article is hypothesis generation. It does not constitute medical advice and does not establish a clinician–patient relationship. Nothing in it should be used to select, request or avoid any treatment. No purinergic receptor antagonist is licensed for the use discussed here, and the research compounds named are not medicines. α -adrenoceptor antagonists are prescription drugs appropriate only under qualified medical supervision.

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